

C6.1 Improving processes and products

Summary

Historically, new materials have been developed through trial and error, experience etc. but as our understanding of the structure of materials and chemical processes has improved we are increasing our ability to manipulate and design new materials. Industry is continually looking to make products that have a better performance and are sustainable to produce. This section also explores the extraction of raw materials and their use in making new products.

Underlying knowledge and understanding

Learners should be familiar with the properties of ceramics, polymers and composites. They should have knowledge of the order of metals and carbon in

the reactivity series. Learners should have met the method of using carbon to obtain metals from metal oxides. They should also be aware that the Earth has limited resources and the benefits of recycling materials.

Common misconceptions

Learners often think that chemical reactions will continue until all the reactants are exhausted. They also think that equilibrium is a static condition.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM6.1i	arithmetic computation, ratio when measuring rates of reaction	M1a, M1c
CM6.1ii	drawing and interpreting appropriate graphs from data to determine rate of reaction	M4b, M4c
CM6.1iii ☒	determining gradients of graphs as a measure of rate of change to determine rate	M4d, M4e
CM6.1iv ☒	proportionality when comparing factors affecting rate of reaction	M1c

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
C6.1a	explain, using the position of carbon in the reactivity series, the principles of industrial processes used to extract metals, including extraction of a non-ferrous metal	the principles of using carbon to extract iron and other metals from their ores	M1a, M1c	WS1.4a	Extraction of copper by heating copper oxide with carbon. (PAG C1)
C6.1b	explain why and how electrolysis is used to extract some metals from their ores		M4b, M4c	WS1.3a, WS1.3b, WS1.3c, WS1.3d, WS1.3e, WS1.3g, WS1.3h, WS1.3i, WS1.4, WS2b	Electrolysis of aqueous sodium chloride solution. (PAG C2) Electrolysis of aqueous copper sulfate solution. (PAG C2)
C6.1c	evaluate alternative biological methods of metal extraction	bacterial and phytoextraction		WS1.1a, WS1.1e	
C6.1d ☒	explain the trade-off between rate of production of a desired product and position of equilibrium in some industrially important processes	the Haber process and Contact process	M4d, M4e	WS1.3f	
C6.1e ☒	interpret graphs of reaction conditions versus rate		M1c	WS1.3e	
C6.1f ☒	explain how the commercially used conditions for an industrial process are related to the availability and cost of raw materials and energy supplies, control of equilibrium position and rate			WS1.1d	
C6.1g ☒	explain the importance of the Haber process in agricultural production			WS1.4a	

C6.1h ☒	compare the industrial production of fertilisers with laboratory syntheses of the same products			WS1.2a, WS1.2b, WS1.2c, WS1.2d, WS1.2e, WS2a, WS2b	Preparation of potassium sulfate or ammonium sulfate using a titration method. (PAG C6)
C6.1i ☒	recall the importance of nitrogen, phosphorus and potassium compounds in agricultural production			WS1.4a	
C6.1j ☒	describe the industrial production of fertilisers as several integrated processes using a variety of raw materials	ammonium nitrate and ammonium sulfate		WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS2a, WS2b	
C6.1k	describe the basic principles in carrying out a life-cycle assessment of a material or product	the use of resources and impact on the environment of all stages of a life-cycle assessment: • making materials for a product from raw materials through to the process used to make the product • the use of the product • transport of the product • the method used for its disposal at the end of its life			

C6.1l	interpret data from a life-cycle assessment of a material or product				
C6.1m	describe a process where a material or product is recycled for a different use, and explain why this is viable			WS1.1f, WS1.1g	
C6.1n	evaluate factors that affect decisions on recycling			WS1.1f, WS1.1g	
C6.1o ☒	describe the composition of some important alloys in relation to their properties and uses	steel, brass, bronze, solder, duralumin			
C6.1p ☒	describe the process of corrosion and the conditions which cause corrosion	iron and other metals			
C6.1q ☒	explain how mitigation of corrosion is achieved by creating a physical barrier to oxygen and water and by sacrificial protection				
C6.1r ☒	compare quantitatively the physical properties of glass and clay ceramics, polymers, composites and metals				
C6.1s ☒	explain how the properties of materials are related to their uses and select appropriate materials given details of the usage required			WS1.1e, WS1.3f	